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Federated Learning for Diabetic Retinopathy Detection in a Multi-center Fundus Screening Network

Sarah Matta,[1] Mariem Ben Hassine,[2] Clément Lecat,[3] Laurent Borderie,[3]
Alexandre Le Guilcher,[3] Pascale Massin,[4] Béatrice Cochener,[5] Mathieu Lamard,[1] and Gwenolé Quellec[6]

***Abstract*— Federated learning (FL) is a machine learning framework that allows remote clients to collaboratively learn a global model while keeping their training data localized. It has emerged as an effective tool to solve the problem of data privacy protection. In particular, in the medical field, it is gaining relevance for achieving collaborative learning while protecting sensitive data. In this work, we demonstrate the feasibility of FL in the development of a deep learning model for screening diabetic retinopathy (DR) in fundus photographs. To this end, we conduct a simulated FL framework using nearly 700,000 fundus photographs collected from OPHDIAT, a French multi-center screening network for detecting DR. We develop two FL algorithms: 1) a cross-center FL algorithm using data distributed across the OPHDIAT centers and 2) a cross-grader FL algorithm using data distributed across the OPHDIAT graders. We explore and assess different FL strategies and compare them to a conventional learning algorithm, namely centralized learning (CL), where all the data is stored in a centralized repository. For the task of referable DR detection, our simulated FL algorithms achieved similar performance to CL, in terms of area under the ROC curve (AUC): AUC = 0.9482 for CL, AUC = 0.9317 for cross-center FL and AUC = 0.9522 for cross-grader FL. Our work indicates that the FL algorithm is a viable and reliable framework that can be applied in a screening network.**

***Clinical relevance*— Given that data sharing is regarded as an essential component of modern medical research, achieving collaborative learning while protecting sensitive data is key.**

I. INTRODUCTION

With the growing advances in Deep Learning (DL) research in the medical field, collaborative multi-center projects have shown an increased interest over the past few years. As DL techniques are data-driven, large and heterogeneous annotated data sets, collected from multiple sites, are desirable for developing robust and externally generalizable DL models intended for widespread clinical use [1]. Through multi-site collaboration, DL models can experience heterogeneous data sources that cover a large spectrum of patient populations, clinical protocols, and data acquisition equipment (i.e. scanners, digital fundus cameras) [2].

Despite the promise of global collaborative training to access a large amount of data, medical data sharing is conflicted to data privacy protection challenges [3]. One major concern is that medical data may include sensitive and private patient information that is commonly prohibited from outsourcing, particularly when full de-identification cannot be ensured [4]. In this context, there are several laws and regulations in place to protect the privacy of clinical data. Namely, the European General Data Protection Regulation (GDPR) [5] and the Health Insurance Portability and Accountability Act (HIPAA) of the United States [6].

Federated Learning (FL), which is a distributed machine learning framework proposed by McMahan et al [7], is considered to be a promising solution to the aforementioned issue. In contrast to traditional centralized learning, FL algorithms learn from decentralized data distributed across different participants (aka clients) [4], [8]. It typically involves a central server which coordinates the training process. It enables collaborative learning amongst many parties while ensuring privacy by keeping locally the training datasets of each client and only sharing the locally trained models with the central server. At each round, at the client level, a DL algorithm is trained using its local data and shares the resulting model with the central server. At the central server level, these models are aggregated into an updated global model [7]. In the next round, the updated global model is redistributed to the clients.

Based on the types of clients, federated learning can be further grouped into two major categories. The first category, cross-device FL, requires a million devices (such as smartphones, wearables, and edge devices) with little data. In contrast, the second category, cross-silo FL, permits a smaller number of collaborative clients with large sample sizes; these participants are often trustworthy businesses or organizations, such as banks or hospitals [9].

Recent studies have investigated the use of cross-silo FL in the medical field. More particularly, this approach was adopted for the COVID-19 pandemic [10], [11]. In ophthalmology, recent studies [12], [13] have explored FL for developing a DL model for retinopathy of prematurity (ROP). The FL framework was also adopted to detect referable Diabetic Retinopathy (DR) using Optical Coherence Tomography (OCT) and OCT-angiography from two different institutions [14].

[1] S. Matta and M. Lamard are with Université de Bretagne Occidentale, Brest, F-29200 France and Inserm, UMR 1101, Brest, F-29200 France. sarah.matta@univ-brest.fr

[2] M. Ben Hassine is with the Faculty of Science and technology, Université Paris-Est Créteil, Créteil, F-94000 France

[3] C. Lecat, L. Borderie and A. Le Guilcher are with Evolucare Technologies, Villers-Bretonneux, F-80800 France.

[4] P. Massin is with Service d'Ophtalmologie, Hôpital Lariboisière, APHP, Paris, F-75475 France.

[5] B. Cochener is with Inserm, UMR 1101, Brest, F-29200 France, Université de Bretagne Occidentale, Brest, F-29200 France and Service d'Ophtalmologie, CHRU Brest, Brest, F-29200 France.

[6] G. Quellec is with Inserm, UMR 1101, Brest, F-29200 France

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In this work, we investigate the FL approach for screening referable DR in fundus photographs in a simulated cross-silo environment. Referable DR, the stage at which the patient should be referred to an ophthalmologist, is defined as moderate or severe non-proliferative DR (NPDR) or proliferative DR, with or without macular edema (ME). In contrast to the aforementioned work which typically uses multiple, independent datasets to simulate a FL framework, we simulated a FL framework using a fundus photographs dataset collected from a single multi-center screening network: OPHDIAT [15]. In OPHDIAT, images from multiple screening centers are centralized on a single server and then randomly assigned to graders from a single pool of graders. This is a unique opportunity to investigate FL: intrinsically, this data can be distributed respectively to the screening center or respectively to the grader. As a result, two factors that affect DL generalizability can be studied: the difference in terms of collected imaging data (screening centers) and the difference of annotation between readers (graders). Two FL algorithms are thus developed: a cross-center FL algorithm, using the OPHDIAT dataset distributed across screening centers, and a cross-grader FL algorithm, using the OPHDIAT dataset distributed across graders. Finally, the performances of the proposed FL algorithms were compared with that of a model trained on a fully centralized dataset.

II. METHODOLOGY

A. Federated Averaging Algorithm

The FL framework is an iterative process divided into federated learning rounds (which are sets of client-server interactions) for developing a global, central DL model, with the constraint of preserving the clients' privacy. It assumes that there is a predefined set of clients, denoted by K. Each k client has a local private data, d_k, which is not shared among other clients neither to the server. Let $t = 1, 2, ..., T$ denote the number of rounds. At the start of each round, a set of clients, m, is selected and the server transmits the current global algorithm state to each of these clients (e.g., the current model parameters). Then, each selected client trains the current model on its local data for a fixed number of epochs (E) and for a predefined minibatch size. The resulting local models, $w_{k,t}$, are then uploaded to the server end for aggregation. Finally, this process is repeated for T rounds [7].

We studied different FL aggregation at the server level. The first strategy, Unweighted Aggregation (UA), calculates an unweighted average of all received model vectors. This fusion method assumes that all clients' models are equally contributing to build the global model. Another strategy, Weighted Aggregation (WA), consists of computing a weighted average of all vectors based on the number of training data of each client. In this approach, a client who has more training data should have more impact on the global model than one with less training data.

We also explored different methods for selecting the m clients at each round. The first one was Random Selection (RS) and the second one was Ordered Selection (OS). For OS, the clients were sorted out based on the number of samples they have. At each round, the clients with the larger number of samples are selected, and the remaining clients are passed to the next round.

B. FL Algorithm Development for DR Detection

For the FL algorithm development, each d_k was split into a training subset (d_k^{train}, 80% of d_k), used to optimize the local weights of each client, and a validation subset ($d_k^{validation}$, 20% of d_k), used to choose the best model. Each client locally trained a DL model ($w_{k,t}$) on d_k^{train} for DR detection. The model was validated on $d_k^{validation}$ using the area under the receiver operating characteristic curve (AUC) as an evaluation metric (AUC$_{k,t}$). After each round, model weights ($w_{k,t}$) and validation metrics (AUC$_{k,t}$) from each client were sent to the server for aggregation. At each round, we compute the mean validation AUC using the m AUC$_{k,t}$ which were sent to the server. When training is done, the global FL model which maximizes the mean validation AUC is selected.

C. Centralized Learning Algorithm

The centralized learning (CL) algorithm is a machine learning framework where data is processed in a centralized location rather than being distributed. In such a setting, all the development data ($D_{dev} = \bigcup_k d_k$) is pooled in a single location and is divided into a training subset (D_{Train}, 80% of D_{dev}) and a validation subset ($D_{Validation}$, 20% of D_{dev}). A single DL model is trained and validated using D_{Train} and $D_{Validation}$, respectively. We use the validation AUC as an evaluation metric to choose the best model.

D. Algorithms Evaluation

The FL algorithm and the CL algorithm were evaluated on the same testing set, referred to as D_{Test}. The performances of both algorithms for detecting referable DR were evaluated using the test AUC. To measure the efficiency of the FL framework, we report the training time per round and the time taken for obtaining the best global model.

III. EXPERIMENTAL SETUP AND DATASETS

A. OPHDIAT screening network

With the aim to enhance screening for DR in France, the Public Assistance Hospitals of Paris (AP-HP) developed OPHDIAT, a telemedical network system comprising a server, screening centers and a reading center. The system operates as follows: in the screening centers, technicians take fundus photographs using non-mydriatic cameras. These photographs are then teletransmitted to a reference center where ophthalmologists grade them. And finally, a generated report is sent to the corresponding center. This report constitutes a basis on which further examinations are commended for patients with referable DR or non-gradable photographs [15]. Thanks to this framework, the regular annual evaluation for DR has improved. OPHDIAT's components can be presented as follows:

- *Screening centers.* To date, the network comprises 46 screening centers located in diabetic wards of hospitals, primary healthcare centers and prisons in the Île-de-France area. Retinal screening was carried out by trained orthoptists or nurses using one of the following 45°digital non-mydriatic cameras: Canon CR-DGI or CR2 (Tokyo, Japan), Topcon TRC-NW6 or TR-NW400 (Rotterdam, The Netherlands). Two retinal digital photographs were taken per eye without pupil dilation, one centered on the posterior pole and the other on the optic disc [16].
- *Server.* All clinical data were transmitted to a medical server where a web-based telemedicine application, OPHCARE, is installed [15].
- *Reading Center.* To date, 25 certified ophthalmologists in the OPHDIAT reading center graded all retinal images according to the International Clinical Diabetic Retinopathy Scale [17].

B. OPHDIAT dataset

The OPHDIAT dataset consists of 164,659 examinations collected from 101,383 patients between 2004 and 2017. For the purpose of this study, eye laterality was determined for each image using our laterality classification algorithm [16]. Next, only gradable images were considered for the development and testing of the DL algorithms: a total of 697,275 images (96,023 patients) were selected. In order to unify their visual aspect, all fundus photographs were cropped and their intensity and hue were normalized following the min-pooling solution [18].

For quality assurance, a subset of 21,054 images from 4,850 patients were annotated by two human readers. We used this subset of better annotated images, referred to as D_{Test}, for testing the FL and the CL algorithms. Then, we ensured that these patients are not present in the development of the algorithms. Thus, a total of 641,917 images (91,173 patients), D_{dev}, was considered for the development of the FL algorithm and the CL algorithm.

For the FL framework, we created a decentralized data from D_{dev} by partitioning it into k local development subsets (d_k) according to screening centers and graders. We removed clients that have less than 8 patients. The OPHDIAT decentralization resulted in $K = 43$ clients across screening centers and $K = 20$ clients across graders.

C. FL Algorithm Development

The cross-center FL algorithm was developed using D_{dev} divided based on screening centers and the cross-grader FL algorithm was developed using D_{dev} divided based on graders. EfficientNet-B5 [19] was used as backbone for all our experiments. Different values for the hyperparameters m and E were explored: $m \in \{5, 15, K\}$ and $E \in \{5, 15\}$. w_t was trained using binary cross-entropy loss with Adam optimizer and a learning rate of 0.001. To select the m clients, RS and OS were investigated. For the aggregation, we investigated UA and WA.

D. CL Algorithm Development

Similar to the FL framework, EfficientNet-B5 [19] was used as backbone. The CNN was trained on D_{Train} using binary cross-entropy loss with Adam optimizer with a learning rate of 0.001.

IV. RESULTS AND DISCUSSION

The CL algorithm achieved an AUC of 0.9482 on the D_{Test} subset for detecting at least moderate DR.

The performances of the cross-center FL algorithm and of the cross-grader FL algorithm on the D_{Test} subset are summarized in Tables I and II, respectively. Both FL algorithms achieved good performances for detecting at least moderate DR with an AUC ranging from 0.9326 to 0.9522 on the D_{Test} subset, which is comparable to the performances obtained by the centralized learning algorithm (AUC: 0.9482).

For the cross-center FL algorithm, the FL strategy using unweighted aggregation and random selection with $E = 5$ and $m = 10$ achieved the best AUC of 0.9373. The best global model was obtained after 7 days of training using one GeForce RTX 2080 graphical processing unit from Nvidia (Santa Clara, USA).

For the cross-grader FL algorithm, the FL strategy using unweighted aggregation and random selection with $E = 15$ and $m = 20$ achieved the best AUC of 0.9522. This global model was obtained after 15 days of training. In comparison, using the same FL strategy, the cross-center FL algorithm achieved an AUC of 0.9317.

We first hypothesized that the cross-grader FL algorithm outperformed the cross-center FL algorithm because its clients have larger training datasets. To this end, we studied how the size of the local training data impacts the performance of each client for the cross-center FL algorithm and the cross-grader FL algorithm. To compare both settings, we used the UA FL strategy with $E = 15$ and $m = K$. We illustrate in Figures 1 and 2 the performances of each client on its local validation data ($d_k^{validation}$) versus the number of training images (size of d_k^{train}). It can be observed that the AUC is not significantly correlated with the size of the client training data ($p = 0.1839$ for the cross-center FL algorithm and $p = 0.4628$ for the cross-grader FL algorithm).

Another potential explanation for the superiority of cross-grader FL, over both CL and cross-center FL, is that inter-grader variability is better taken into account in this scenario. The best aggregation strategy, namely unweighted aggregation (UA), gives equal weight to all graders, leading to a more universal model.

In this study, we presented a proof of concept for the application of FL in the development of DL models for DR screening in fundus photographs. Our results suggest that annotations have more impact on the DL generalizability than the difference existing in images due to using different acquisition protocols. This work has shown that FL has achieved comparable performances to centralized learning. In summary, FL is a promising approach to preserve privacy in collaborative projects in health research.

TABLE I

PERFORMANCES OF THE CROSS-CENTER FL ALGORITHM

FL strategy	AUC	Time per round (hours)	optimal time (days)
UA, $E = 15, m = 43$	0.9317	78	15
UA, $E = 5, m = 43$	0.9297	29	7
RS, UA, $E = 5, m = 10$	0.9373	22	7
RS, UA, $E = 5, m = 5$	0.9326	5	2
RS,WA, $E = 5, m = 5$	0.9370	6	3
OS,UA, $E = 5, m = 5$	0.9333	5	2.5

TABLE II

PERFORMANCES OF THE CROSS-GRADER FL ALGORITHM

FL strategy	AUC	Time per round (hours)	optimal time (days)
UA, $E = 15, m = 20$	0.9522	78	15
UA, $E = 5, m = 20$	0.9470	24	6
RS, UA, $E = 5, m = 5$	0.9371	8	1.7
RS, WA, $E = 5, m = 5$	0.9442	8	3
OS, UA, $E = 5, m = 5$	0.9424	4	0.3

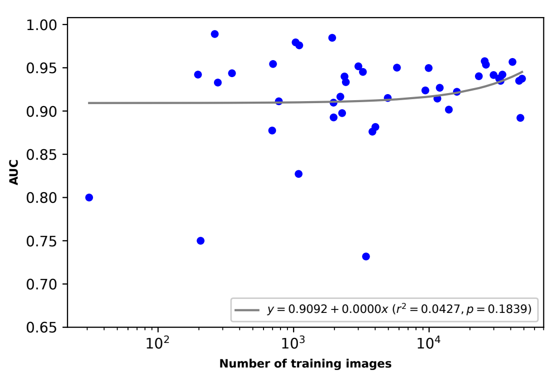

Fig. 1. AUC versus the number of training images of each client using the cross-center FL algorithm. Each client is represented by a circle. r denotes Pearson's correlation coefficient.

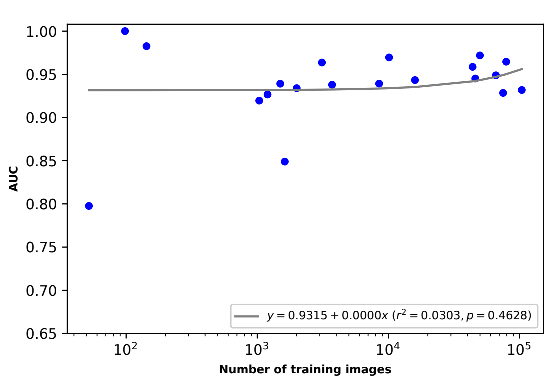

Fig. 2. AUC versus the number of training images for each client using the cross-grader FL algorithm. Each client is represented by a circle.

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